

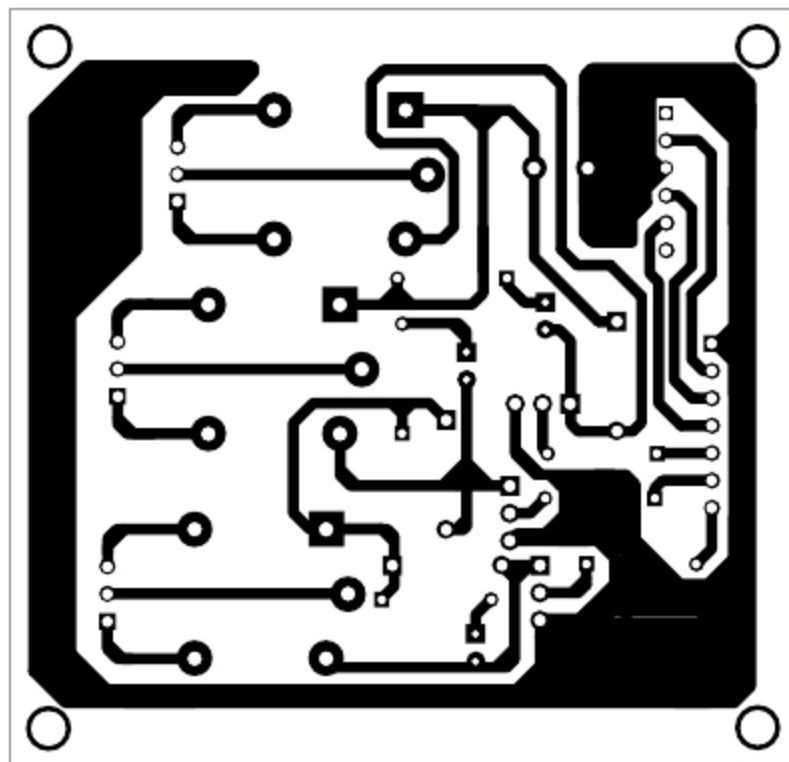


Fig. 4: Actual-size PCB layout of home automation system

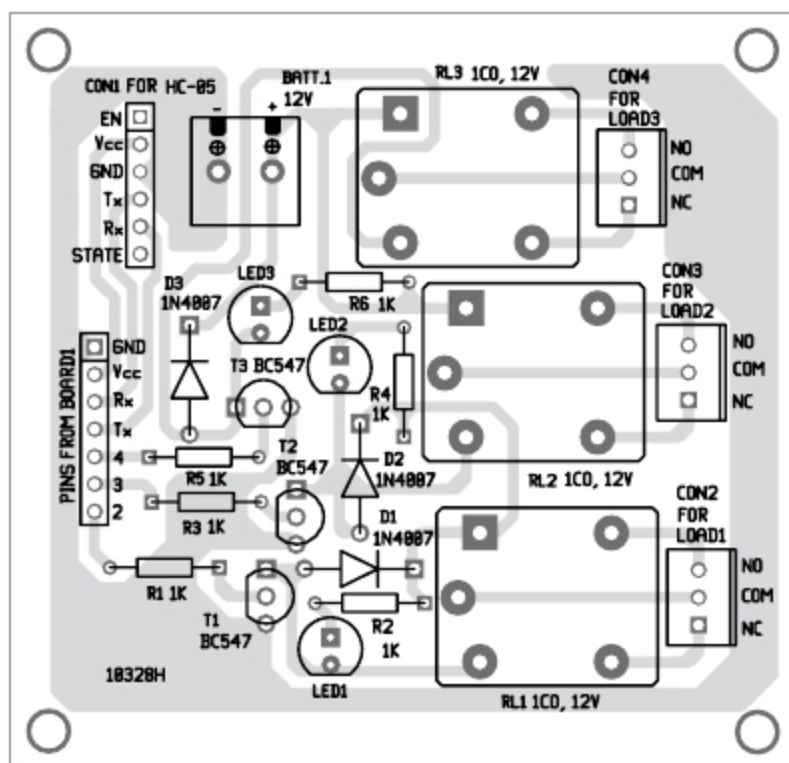


Fig. 5: Components layout for the PCB

electrical load, and the other terminal of the load to neutral (N) line of 220V AC mains power supply. Connect live (L) line 220V to COM terminal of each relay.

Construction and testing

An actual-size PCB layout for the home automation system using Arduino Uno through Android app is shown in Fig. 4 and its components layout

EFY Note
The source code of this project is available for free download at source.efymag.com



Fig. 6: QR code for Arduino source code



Fig. 7: QR code for Android app



Fig. 8: Android app on smartphone

in Fig. 5. Assemble the components on the PCB as per the circuit diagram.

Then, connect the power supply to Arduino Uno either through its USB cable or 12V battery. Also, connect 12V power to all three relays, as shown in the circuit diagram.

Upload Arduino source code (HomeAutomation.ino) to Arduino Uno. The source code can be downloaded by scanning the QR code given in Fig. 6. Next, pair Bluetooth module HC-05 with Bluetooth of your smartphone.

Download Android app (HomeAutomation.apk) using the QR code

given in Fig. 7. Install it on your Android smartphone. Next, open the app and pair Bluetooth module HC-05 with the Bluetooth of your smartphone. You will get the app on your phone as shown in Fig. 8. Now you may click on the buttons to control the appliances. **EFY**



Kartik Gudali is an electronics enthusiast and RC hobbyist. Currently, he is working on the development of Android, Arduino, Raspberry Pi and Amazon Alexa



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-Henry Ford

